ROS2

DAY 6 – Manual 4

Parameters – using YAML

Goal of this manual

Students will learn:

- What a parameter YAML file is
- How ROS 2 loads parameters from YAML
- How to structure YAML for one node or multiple nodes
- How to launch a node with YAML (without using a launch file)
- How to verify that parameters are loaded
- How to override parameters from YAML + command line

This manual does **NOT** use launch files (covered in Day 07).

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1. Understanding YAML Parameter Files

ROS 2 allows parameters to be stored in a clean, modular **YAML** file.
YAML files make parameter management easier for:

- Clean configuration
- Reusability
- Storing many parameters
- Sharing configurations between team members

A parameter YAML file has 3 layers:

```
<namespace>:
  <node_name>:
    ros__parameters:
      <param_name>: <value>
```

Example:

```
number_param_node:
  ros__parameters:
    number_param: 22
```

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2. Create a YAML File

Step 1 — Open a new terminal

```
cd DAY-06/ros2_ws/src/param_pkg
mkdir config
cd config
nano params.yaml
```

Paste the following:

```
number_param_node:
  ros__parameters:
    number_param: 22
```

Save:

- Ctrl + O → Enter
- Ctrl + X

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3. Rebuild the Workspace

Open a new terminal:

```
cd DAY-06/ros2_ws
colcon build
source install/setup.bash
```

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4. Run Node With YAML Parameters

Open **Terminal 1**:

```
ros2 run param_pkg param_node --ros-args --params-file
src/param_pkg/config/params.yaml
```

Expected output:

```
[INFO] Node started. Publishing number_param every second.
[INFO] Published: 22
```

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5. Verify Parameters Loaded Correctly

Open **Terminal 2**:

```
ros2 param get /param_node number_param
```

Output should be:

```
Integer value is: 22
```

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6. Overriding YAML Parameter at Runtime

While the node is running, open **Terminal 3**:

```
ros2 param set /param_node number_param 123
```

Effect in Terminal 1:

```
[INFO] Parameter updated! New number_param = 123
[INFO] Published: 123
```

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7. Combining YAML + Command-Line Override

Start the node using YAML **but override one parameter**:

```
ros2 run param_pkg param_node \
  --ros-args --params-file src/param_pkg/config/params.yaml \
  -p number_param:=77
```

Result:

```
[INFO] Published: 77
```

Command-line arguments **always win** over YAML.

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8. Full YAML Structure for Multiple Parameters

Example file:

```
param_node:
  ros__parameters:
    number_param: 10
    speed: 5.5
```

```
    enabled: true
```

Parameter types supported:

- integer
- float
- string
- boolean
- lists

Example lists:

```
param_node:
  ros__parameters:
    gains: [0.1, 0.01, 0.5]
```

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9. YAML for Multiple Nodes

You can configure different nodes in one file:

```
number_param_node:
  ros__parameters:
    number_param: 10

other_node:
  ros__parameters:
    threshold: 0.8
```

Or structured under a namespace:

```
my_robot:
  number_param_node:
    ros__parameters:
      number_param: 33

  controller_node:
    ros__parameters:
      kp: 2.5
      ki: 0.3
```
